

Checkpoint 1

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ThreadCreate:

```
chris@Chris MINGW64 /d/NTHU FILES/SEMESTER 6/OS/ppc1
$ make clean
rm: cannot remove '*.lnk': No such file or directory
Makefile:25: recipe for target 'clean' failed
make: *** [clean] Error 1

chris@Chris MINGW64 /d/NTHU FILES/SEMESTER 6/OS/ppc1
$ make
sdcc -c testcoop.c
sdcc -c cooperative.c
cooperative.c:26:1: warning: multi-line comment
   26 | // #define SAVESTATE          \
      | ^
cooperative.c:42:1: warning: multi-line comment
   42 | // #define RESTORESTATE      \
      | ^
cooperative.c:274: warning 85: in function ThreadCreate unreferenced function argument : 'fp'
sdcc -o testcoop.hex testcoop.rel cooperative.rel
```

Functions:

Area	Addr	Size	Decimal Bytes (Attributes)
-----	----	----	-----
CSEG	0000004F	0000019C =	412. bytes (REL,CON,CODE)
	Value	Global	Global Defined In Module
	-----	-----	-----
C:	0000004F	_Producer	testcoop
C:	00000072	_Consumer	testcoop
C:	00000098	_main	testcoop
C:	000000A4	__sdcc_gsinit_startup	testcoop
C:	000000A8	__mcs51_genRAMCLEAR	testcoop
C:	000000A9	__mcs51_genXINIT	testcoop
C:	000000AA	__mcs51_genXRAMCLEAR	testcoop
C:	000000AB	_Bootstrap	cooperative
C:	000000C9	_ThreadCreate	cooperative
C:	00000147	_ThreadYield	cooperative
C:	00000199	_ThreadExit	cooperative

Variables:

Area	Addr	Size	Decimal	Bytes	(Attributes)
-----	----	----	-----	-----	-----
. .ABS.	00000000	00000000 =		0.	bytes (ABS,CON)
Value	Global	Global Defined In Module			
-----	-----	-----			
00000000	__ABS.	cooperative			
00000030	_data_ready	testcoop			
00000031	_shared_buf	testcoop			
00000032	_curr_char	testcoop			
00000038	_active_thread	cooperative			
00000039	_thread_mask	cooperative			
0000003A	_saved_sp	cooperative			
0000003E	_temp_tid	cooperative			
0000003F	_sp_temp	cooperative			
00000080	_P0	cooperative			
00000080	_P0_0	cooperative			
00000081	_P0_1	cooperative			
00000081	_SP	cooperative			
00000082	_DPL	cooperative			
00000082	_P0_2	cooperative			
00000083	_DPH	cooperative			
00000083	_P0_3	cooperative			
00000084	_P0_4	cooperative			
00000085	_P0_5	cooperative			
00000086	_P0_6	cooperative			
00000087	_P0_7	cooperative			

Main:

System Clock (MHz)11.05921000Update Freq.

SBUF

R/O

W/O

TH0

TL0

R7

0x00

B

0x00

0x00

0x00

0x00

0x00

R6

0x00

ACC

0x00

R5

0x00

PSW

0x00

R4

0x00

IP

0x00

R3

0x00

IE

0x00

R2

0x00

PCON

0x00

R1

0x3A

DPH

0x00

R0

0x3A

DPL

0x00

SP

0x3F

RXD

TXD

1

1

TMOD

0x00

TCON

0x00

pins

bits

TH1

TL1

0xFF

0xFF

P3

0x00

0x00

0xFF

0xFF

P2

0x00

0x00

0xFF

0xFF

P1

0x00

0x00

0xFF

0xFF

P0

0x00

0x00

PC

0x0098

PSW

0

0

0

0

0

0

0

0

0

0

Data Memory

addr

0x00

0x00

value

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
00	3A	3A	00	00	00	00	00	00	00	B4	00	00	00	03	00	02
10	00	00	00	00	00	00	00	00	00	00	3A	00	00	00	00	08
20	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
30	01	4B	4C	00	00	00	00	00	00	01	46	56	00	00	00	09
40	98	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
50	92	00	00	01	00	00	09	00	00	00	00	00	00	00	00	00
60	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
70	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00

Remove All Breakpoints

RST

Step

Run

New

Load

Save

CPY

Paste

BP

Time: 115us - Instructions: 70

0067|

MOV A, 32H

0069|

ADD A, #0A5H

006B|

JNC 0E5H

006D|

MOV 32H, #41H

0070|

SJMP 0E0H

0072*

MOV 89H, #20H

0075|

MOV 8DH, #0FAH

0078|

MOV 98H, #50H

007B|

SETB 8EH

007D|

MOV A, 30H

007F|

JNZ 05H

0081|

LCALL 0147H

0084|

SJMP 0F7H

0086|

MOV 99H, 31H

0089|

MOV 30H, #00H

008C|

JB 99H, 05H

008F|

LCALL 0147H

0092|

SJMP 0F8H

0094|

CLR 99H

0096|

SJMP 0E5H

0098*

MOV 30H, #00H

Producer:

System Clock (MHz)11.05921000Update Freq.

SBUF

R/O

W/O

TH0

TL0

R7

0x00

B

0x00

0x00

0x00

0x00

0x00

R6

0x02

ACC

0x3B

R5

0x00

PSW

0x09

R4

0x03

IP

0x00

R3

0x00

IE

0x00

R2

0x00

PCON

0x00

R1

0x00

DPH

0x00

R0

0x3B

DPL

0x01

SP

0x3F

RXD

TXD

1

1

TMOD

0x00

TCON

0x00

pins

bits

TH1

TL1

0xFF

0xFF

P3

0x00

0x00

0xFF

0xFF

P2

0x00

0x00

0xFF

0xFF

P1

0x00

0x00

0xFF

0xFF

P0

0x00

0x00

PC

0x004F

PSW

0

0

0

0

1

0

0

1

Data Memory

addr

0x00

0x00

value

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
00	3A	3A	00	00	01	00	00	10	3B	00	00	00	03	00	02	00
10	00	00	00	00	00	00	00	00	00	3A	00	00	00	00	08	00
20	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
30	00	4B	4C	00	00	00	00	00	00	03	46	56	00	00	01	41
40	A1	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
50	72	00	00	00	00	00	09	00	00	00	00	00	00	00	00	00
60	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
70	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00

Remove All Breakpoints

RST

Step

Run

New

Load

Save

CPY

Paste

BP

Time: 251us - Instructions: 159

004F*

MOV 32H, #41H

0052|

MOV A, #01H

0054|

CJNE A, 30H, 05H

0057|

LCALL 0147H

005A|

SJMP 0F6H

005C|

MOV 31H, 32H

005F|

MOV 30H, #01H

0062|

MOV A, 32H

0064|

INC A

0065|

MOV 32H, A

0067|

MOV A, 32H

0069|

ADD A, #0A5H

006B|

JNC 0E5H

006D|

MOV 32H, #41H

0070|

SJMP 0E0H

0072*

MOV 89H, #20H

0075|

MOV 8DH, #0FAH

0078|

MOV 98H, #50H

007B|

SETB 8EH

007D|

MOV A, 30H

007F|

JNZ 05H

Consumer:

System Clock (MHz)11.05921000Update Freq.

SBUF

R/O

W/O

TH0

TL0

R7

0x00

B

0x00

0x00

0x00

0x00

0x00

R6

0x02

ACC

0x00

R5

0x00

PSW

0x08

R4

0x03

IP

0x00

R3

0x00

IE

0x00

R2

0x00

PCON

0x00

R1

0x3B

DPH

0x00

R0

0x3A

DPL

0x00

SP

0x4F

RXD

TXD

1

1

TMOD

0x00

TCON

0x00

pins

bits

TH1

TL1

PC

0x0072

PSW

0

0

0

0

1

0

0

0

0xFF

0xFF

P3

0x00

0x00

0xFF

0xFF

P2

0x00

0x00

0xFF

0xFF

P1

0x00

0x00

0xFF

0xFF

P0

0x00

0x00

Data Memory

addr

0x00

0x00

value

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
00	3A	3A	00	00	01	00	00	00	00	3A	3B	00	00	03	00	02
10	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
20	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
30	01	41	42	00	00	00	00	00	01	03	46	56	00	00	01	41
40	5A	00	00	01	00	01	09	00	00	00	00	00	00	00	00	00
50	72	00	00	00	00	00	09	00	00	00	00	00	00	00	00	00
60	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
70	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00

Remove All Breakpoints

RSTStepRunNewLoadSaveCPYPasteBP

Time: 345us - Instructions: 219

004F*MOV32H,#41H

0052|MOV A,#01H

0054|CJNE A,30H,05H

0057|LCALL 0147H

005A|SJMP 0F6H

005C|MOV 31H,32H

005F|MOV 30H,#01H

0062|MOV A,32H

0064|INC A

0065|MOV 32H,A

0067|MOV A,32H

0069|ADD A,#0A5H

006B|JNC 0E5H

006D|MOV 32H,#41H

0070|SJMP 0E0H

0072*MOV 89H,#20H

0075|MOV 8DH,#0FAH

0078|MOV 98H,#50H

007B|SETB 8EH

007D|MOV A,30H

007F|JNZ 05H

Output:

U

No Parity

8-bit UART @ 4800 Baud

Rx

QRSTUVWXYZABCDEFGHIJKLMNO

Rx Reset

Tx

Tx Send

0.0 V

input

11111111

ADC

MAX

MIN

Motor Enabled

0000

Explanation:

Implementation Overview

To implement the producer-consumer model, I allocated shared variables in the 0x30 memory range. The Producer runs inside the main thread (Thread 0), continually generating characters and yielding when the buffer is full. The Consumer is dynamically spawned as Thread 1, fetching data, sending it to the serial port, and yielding while waiting for the transmission to finish. I used a bitmap to track active threads and allocated exactly 16 bytes (0x10) for each thread's stack, with the base starting at 0x3F.

Before ThreadCreate

I referenced the map file and set a breakpoint at 007EH, right before the LCALL to the ThreadCreate function. As shown in the screenshot, the stack pointer (SP) is currently at 46H. When the call occurs the LCALL pushes a 2-byte return address, incrementing the stack. Once inside the ThreadCreate, the OS calculates the new thread's dedicated stack space using the formula $0x3F + (\text{ThreadID} * 16)$. For Thread 1, it temporarily shifts the SP to 4FH to safely push the initial thread context (cleared registers and PSW) without corrupting Thread 0's memory.

How I know the Producer is running

The simulator paused at the address 0073H, which matches the _Producer function in my map file. I can confirm the Producer is running by observing the PSW register, which reads 00H. This means the register bank selection bits (RS1 and RS0) are 00, verifying that the CPU is utilizing Register Bank 0. This perfectly aligns with my design, where the Producer runs inside the main thread (Thread 0). Additionally, the SP is at 3FH, matching Thread 0's stack.

How I know the Consumer is running

Execution stopped at 00A5H, which corresponds to the _Consumer function. I know the Consumer is actively running because a context switch has clearly occurred: the PSW register now reads 08H. The RS0 bit has flipped to 1, confirming the microcontroller is now using Register Bank 1 (which was assigned to Thread 1). Furthermore, the SP is now at 4FH, providing execution is safely isolated within the specific 16-byte memory boundary allocated for the Consumer thread.